

Quiz 2 Class 1 Solutions

Computer Security Fundamentals Spring 2026

Instructions: Answer the questions, **closed-book, closed-notes**. Time: 10 minutes

1. (**F**) To achieve the same level of security, symmetric encryption needs to have longer keys than asymmetric encryption. (True/False, 2 pts)
2. (**D**) Alice wants to send an encrypted message to Bob with asymmetric encryption. She needs to: (2 pts)
- A. Encrypt the message with her secret key
 - B. Encrypt the message with her public key
 - C. Encrypt the message with Bob's secret key
 - D. Encrypt the message with Bob's public key
3. (**A**) Public Key Infrastructure (PKI) is necessary for secure asymmetric encryption because: (2 pts)
- A. There's no way to authenticate a public key without PKI
 - B. There's no way to exchange a secret key without PKI
 - C. PKI is necessary for public and secret key pair generation
 - D. PKI is necessary for switching to asymmetric encryption from symmetric encryption
4. (Short Answer, 4 pts) Alice and Bob are your classmates. One day Bob told you that he is trying to express his love to Alice, but he is too afraid to speak out openly and will only send the hash of his message to Alice, with a well-known publicly available hash algorithm. You doubted because you know hashing is one-way. However, a few days later Bob showed up with Alice, and said that Alice had learnt the message, they are now lovers. Can you explain what probably happened with your knowledge of cryptography and passwords?

Solution: The most probable explanation is that Alice is attempting something similar to a dictionary attack. She guessed a few possible short messages like "iloveyou" "I love you", calculated their hash, compared them against the hash send by Bob and found a match.

Not necessary in your answer: Of course, this only works if Alice knows Bob's intention.